

Before ISRO makes us fly

Gaganyaan's real test begins now with G1

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Spaceflight loves a launch, and so do we, as the audience. It gives us fire, noise, countdowns and that moment when several hundred tonnes of very expensive engineering wins its fight with gravity in a triumphant moment.

However, spaceflight becomes different when humans are involved in it. That is why the most consequential Gaganyaan flight may be the one with no astronauts aboard, and it's coming up soon.

India's first uncrewed experimental Gaganyaan mission, targeted for the fourth quarter of 2026, is expected to carry Vyommitra, ISRO's half-humanoid robot. It precedes two further uncrewed missions and the first crewed orbital flight, which is targeted for 2027.

Strip away the symbolism and G1 ultimately boils down to one word – trust. Can India launch a human-rated spacecraft, operate it in orbit, bring it back through the atmosphere, slow a multi-tonne capsule under parachutes, drop it into the sea and recover it as though lives depended on it?

Hardware vs humans

ISRO already knows how to send complex machines into space. Chandrayaan-3 landed near the lunar south polar region. Aditya-L1 operates around the Sun-Earth L1 point. LVM3 has flown major national and commercial missions.



But a satellite does not need breathable air. It does not need an escape route. It does not experience injury-inducing acceleration, require a comfortable temperature or need rescuing from a capsule bobbing in the ocean. Human-rating therefore changes the engineering philosophy.

The Human Rated LVM3, or HLVM3, is derived from LVM3, but the standard is no longer simply "reach the intended orbit". In August, the government said development and ground testing of all HLVM3 propulsion stages and structures had been completed. The CE20 cryogenic engine used by LVM3 has also completed human-rating qualification requirements.

Then there is the Orbital Module: the Crew Module that eventually houses the astronauts, and the unpressurised Service Module that supplies propulsion, power and other support while in orbit. ISRO says propulsion systems for both have been developed, tested and qualified. The Service Module system has been put through nominal and off-nominal scenarios because spacecraft, annoyingly, do not always fail according to PowerPoint. That last point is the essence of Gaganyaan.

Build for the bad day

A crewed spacecraft cannot merely be very good when everything goes right. It has to remain survivable when things go wrong.

Gaganyaan's Crew Escape System is designed to rapidly pull the crew module away from the launch vehicle during an emergency. ISRO's